THOLKAPPIAN MURUGESAN

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EDUCATION

Indian Institute of Information Technology (IIIT) Sri City 2024-2028

B.Tech in Computer Science and Engineering CGPA: 8.71

WORK EXPERIENCE

Full-Stack Developer Intern | Forge Your Peak* **May 2026 - present**

- Architecting a custom EdTech platform using ASP.NET Core and React to replace legacy third-party infrastructure and eliminate 100% platform transaction fees and reduce monthly operation costs by 30%.
- Designing a normalized PostgreSQL database schema with Entity Framework Core, enforcing strict JWT Claims-Based Authorization to securely manage data for active students.
- Prototyping an adaptive-bitrate HLS video streaming pipeline using Cloudflare R2, aiming to cut bandwidth egress costs by 100% and improve global video load times by up to 40%
- Building automated C# migration scripts to safely parse legacy platform CSV exports and bulk-seed relational database tables with zero data loss.

PROJECTS

ProMap | React, Node.js, Express, MongoDB | Live Demo🔗 | Github🔗

- Engineered a secure RESTful API with 15+ endpoints using Express and JWT/bcrypt for secure user authentication and routing.
- Built a responsive React frontend utilizing React [Context API] for global state management, seamlessly handling 2 core user flows like board creation and task assignment.
- Designed a scalable MongoDB database with 4 optimized schemas (Users, Boards, Tasks, etc.), enabling efficient data retrieval and one-to-many relationship mapping.

Flow | React, Node.js, Express, MongoDB | Live Demo🔗 | Github🔗

- Engineered a 3D production tracking platform utilizing a Vite-powered React frontend and a Node.js/Express backend to orchestrate complex asset lifecycles sequentially across multiple rendering stages.
- Implemented robust Role-Based Access Control (RBAC) and JWT authentication middleware, securely isolating API routes and operational privileges among platform owners, administrators, and team members.
- Designed a responsive asset management UI featuring dynamic state tracking and automated deadline highlighting, utilizing Axios to ensure optimized, real-time client-server data synchronization.

CricTrac | PyTorch, OpenCV, RoboFlow | Github🔗

- Curated and annotated a custom dataset of 1,000+ frames using Roboflow to train a highly specialized object detection model for sports analytics.
- Designed a multi-model computer vision pipeline utilizing PyTorch and OpenCV, successfully achieving consistent, frame-by-frame bat and ball tracking across complex video sequences.
- Prototyping spatial coordinate logic to synchronize frame detection data and calculate real-time batting metrics (e.g., bat speed, impact angle) from raw video telemetry.

TECHNICAL SKILLS

- Languages: C, C++, Java, Python, JavaScript, TypeScript, HTML/CSS.
- Backend & Databases: Node.js, Express.js, MongoDB, PostgreSQL RESTful APIs.
- Frontend & Tools: React.js, Git, Vercel, Render, Postman.
- Core Competencies: Data Structures & Algorithms, Object-Oriented Design, Problem Solving, Full-Stack Architecture.

ACHIEVEMENTS

- Awards: 1st Place at Innoventures (Intracollege Hackathon).
- Algorithms: Solving NeetCode 150 & Blind 75 (Focus: DP, Graphs, Backtracking).